

SAND POINT 1 ENE, AK, USA

WMO: 703150

Lat: 55.3473N

Lon: 160.4660W

Elev: 73

StdP: 100.45

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 25630

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-11.7	-9.6	-15.8	1.0	-10.9	-13.8	1.1	-8.7	12.8	1.3	10.4	-1.5	5.9	N/A	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	16.8	13.6	15.4	12.8	14.3	12.1	14.1	16.0	13.3	14.9	12.7	14.0	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	9.6	14.6	12.6	9.2	13.8	11.9	8.7	13.2	39.7	16.2	37.6	14.8	35.9	14.1	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.6	8.2	DB	-11.6	19.7	2.6	2.4	-13.5	21.4	-15.0	22.8	-16.5	24.1	-18.4	25.9	(4)
(5)			WB	-12.1	15.8	2.5	1.6	-13.9	16.9	-15.4	17.8	-16.8	18.7	-18.6	19.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	0.3	0.3	-0.5	2.0	5.1	8.5	11.0	11.8	9.6	6.2	2.9	0.9	(6)
(7)		DBStd	5.16	3.70	3.64	3.99	2.78	2.17	2.13	1.83	1.75	2.02	2.43	3.10	3.57	(7)
(8)		HDD10.0	1998	300	271	326	238	151	54	10	3	31	119	214	281	(8)
(9)		HDD18.3	4912	558	504	584	488	409	294	229	202	263	377	464	539	(9)
(10)		CDD10.0	129	0	0	0	0	1	10	40	61	18	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.4	4.0	4.0	3.4	3.7	3.4	3.0	2.2	2.5	3.0	3.6	3.9	4.2	(14)
(15)	Precipitation	PrecAvg	1051	98	90	72	58	66	75	93	76	113	97	102	111	(15)
(16)		PrecMax	1430	160	185	148	95	125	125	146	113	183	166	167	239	(16)
(17)		PrecMin	789	40	29	21	19	17	32	37	28	75	49	38	41	(17)
(18)		PrecStd	155	32	37	33	21	25	23	27	21	27	31	33	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.9	6.5	8.8	12.8	18.3	18.4	18.8	15.7	11.9	9.4	7.5	(19)
(20)			MCWB	5.0	4.9	4.2	5.8	8.0	13.2	14.4	14.9	12.5	9.8	8.3	7.2	(20)
(21)		2%	DB	5.3	5.2	5.3	6.9	10.6	15.0	16.4	17.0	14.1	10.9	8.7	6.3	(21)
(22)			MCWB	4.5	4.5	4.1	4.5	7.0	11.1	13.3	14.2	12.2	9.4	8.0	5.9	(22)
(23)		5%	DB	4.7	4.8	4.7	5.8	9.1	13.3	15.0	15.6	13.2	10.2	7.9	5.6	(23)
(24)			MCWB	3.9	4.0	3.5	4.1	6.4	10.2	12.6	13.2	11.6	8.9	7.1	5.0	(24)
(25)		10%	DB	4.2	4.3	3.9	5.1	8.0	11.9	13.8	14.5	12.4	9.5	6.9	4.9	(25)
(26)			MCWB	3.3	3.5	2.7	3.9	6.0	9.6	11.8	12.5	11.0	8.2	5.9	4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	5.5	5.2	6.3	8.8	13.7	15.0	15.7	14.0	10.8	9.1	7.5	(27)
(28)			MCDWB	5.8	5.6	5.5	8.0	11.2	18.1	17.2	18.0	14.4	11.0	9.5	7.8	(28)
(29)		2%	WB	4.9	5.0	4.7	5.2	7.8	11.8	13.7	14.6	13.0	10.0	8.4	6.2	(29)
(30)			MCDWB	5.1	5.1	5.1	6.1	9.8	14.5	15.9	16.5	13.5	10.5	8.6	6.2	(30)
(31)		5%	WB	4.1	4.4	3.9	4.7	7.0	10.7	13.0	13.6	12.1	9.2	7.3	5.2	(31)
(32)			MCDWB	4.4	4.7	4.6	5.6	8.6	12.9	14.7	15.1	12.7	9.9	7.8	5.4	(32)
(33)		10%	WB	3.5	3.8	2.9	4.1	6.4	9.9	12.3	12.9	11.4	8.4	6.4	4.2	(33)
(34)			MCDWB	4.0	4.2	3.8	5.0	7.7	11.8	13.6	14.2	12.2	9.3	7.0	4.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.3	3.5	4.2	4.1	4.5	4.9	4.5	4.6	4.4	3.7	3.6	3.3	(35)
(36)			MCDBR	3.1	3.2	4.3	4.8	6.3	8.0	6.3	6.2	5.1	3.9	3.7	3.4	(36)
(37)			MCWBR	3.1	3.2	3.5	3.6	4.0	5.0	4.2	4.0	3.6	3.4	3.7	3.7	(37)
(38)		5% WB	MCDWB	3.0	3.0	3.8	4.5	5.1	7.1	5.8	5.4	4.1	3.5	3.7	3.6	(38)
(39)			MCWBR	3.1	3.3	3.4	3.6	3.5	4.6	4.1	3.8	3.5	3.3	3.9	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.278	0.326	0.349	0.380	0.385	0.386	0.370	0.379	0.362	0.342	0.323	0.283	(40)	
(41)		taud	2.400	2.312	2.260	2.237	2.293	2.370	2.482	2.437	2.417	2.330	2.184	2.339	(41)	
(42)		Ebn at Noon	675	744	813	838	857	859	869	836	798	711	558	565	(42)	
(43)		Edh at Noon	49	77	102	120	121	113	100	98	88	75	60	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.51	1.04	2.37	3.62	4.45	4.78	4.27	3.54	2.56	1.45	0.65	0.35	(44)	
(45)		RadStd	0.07	0.12	0.37	0.22	0.54	0.55	0.46	0.33	0.22	0.11	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page